



# Continuous Random Variables and Probability Distributions



# Chapter Goals

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**After completing this chapter, you should be able to:**

- Explain the difference between a discrete and a continuous random variable
- Describe the characteristics of the uniform and normal distributions
- Translate normal distribution problems into standardized normal distribution problems
- Find probabilities using a normal distribution table



# Chapter Goals

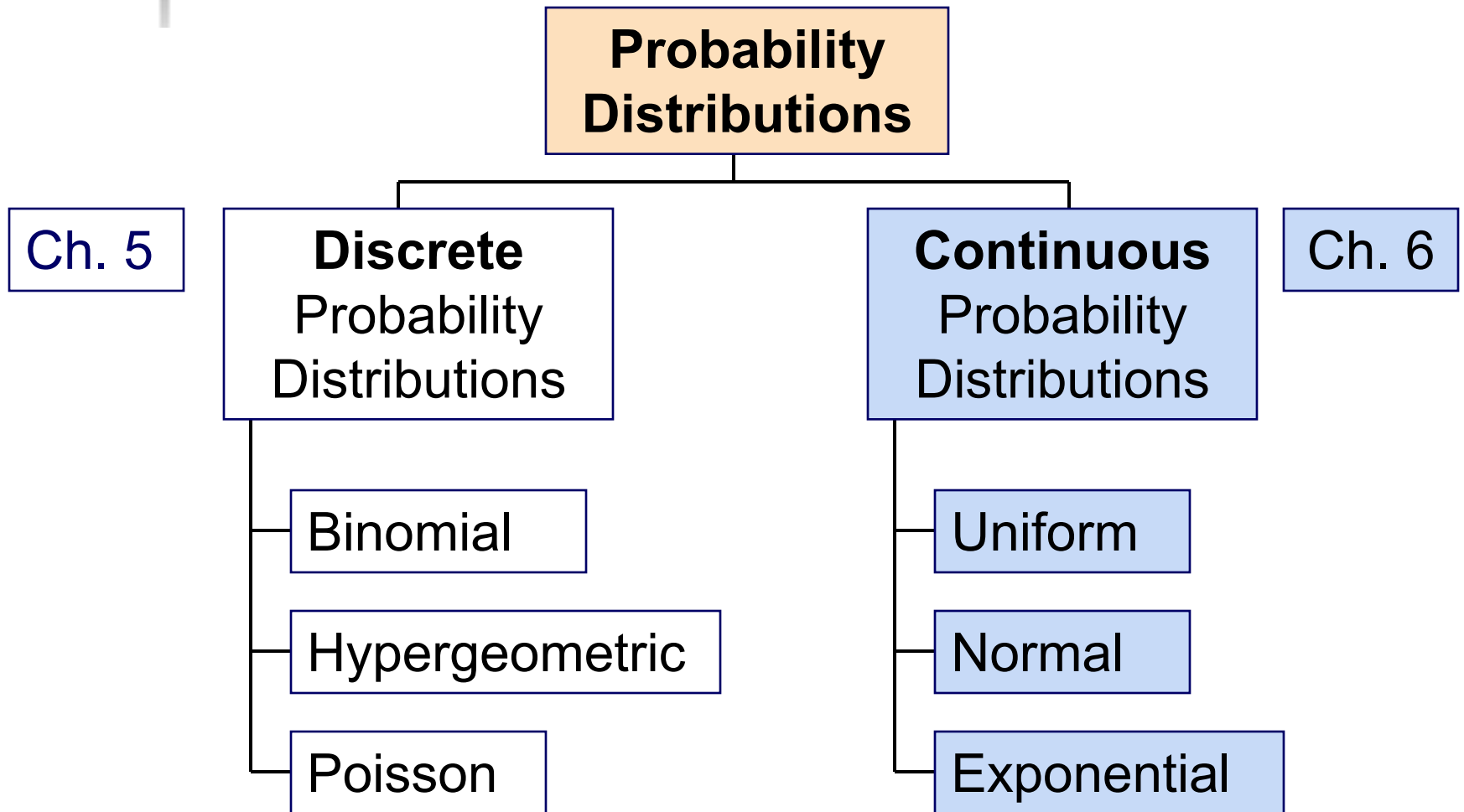
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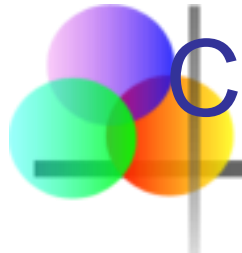
**After completing this chapter, you should be able to:**

- Evaluate the normality assumption
- Use the normal approximation to the binomial distribution
- Recognize when to apply the exponential distribution



# Probability Distributions





# Continuous Probability Distributions

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- A **continuous random variable** is a variable that can assume any value in an interval
  - thickness of an item
  - time required to complete a task
  - temperature of a solution
  - height, in inches
- These can potentially take on any value, depending only on the ability to measure accurately.



# Cumulative Distribution Function

- The **cumulative distribution function**,  $F(x)$ , for a continuous random variable  $X$  expresses the probability that  $X$  does not exceed the value of  $x$

$$F(x) = P(X \leq x)$$

- Let  $a$  and  $b$  be two possible values of  $X$ , with  $a < b$ . The probability that  $X$  lies between  $a$  and  $b$  is

$$P(a < X < b) = F(b) - F(a)$$



# Probability Density Function

The **probability density function**,  $f(x)$ , of random variable  $X$  has the following properties:

1.  $f(x) > 0$  for all values of  $x$
2. The area under the probability density function  $f(x)$  over all values of the random variable  $X$  is equal to 1.0
3. The probability that  $X$  lies between two values is the area under the density function graph between the two values
4. The **cumulative density function**  $F(x_0)$  is the area under the probability density function  $f(x)$  from the minimum  $x$  value up to  $x_0$

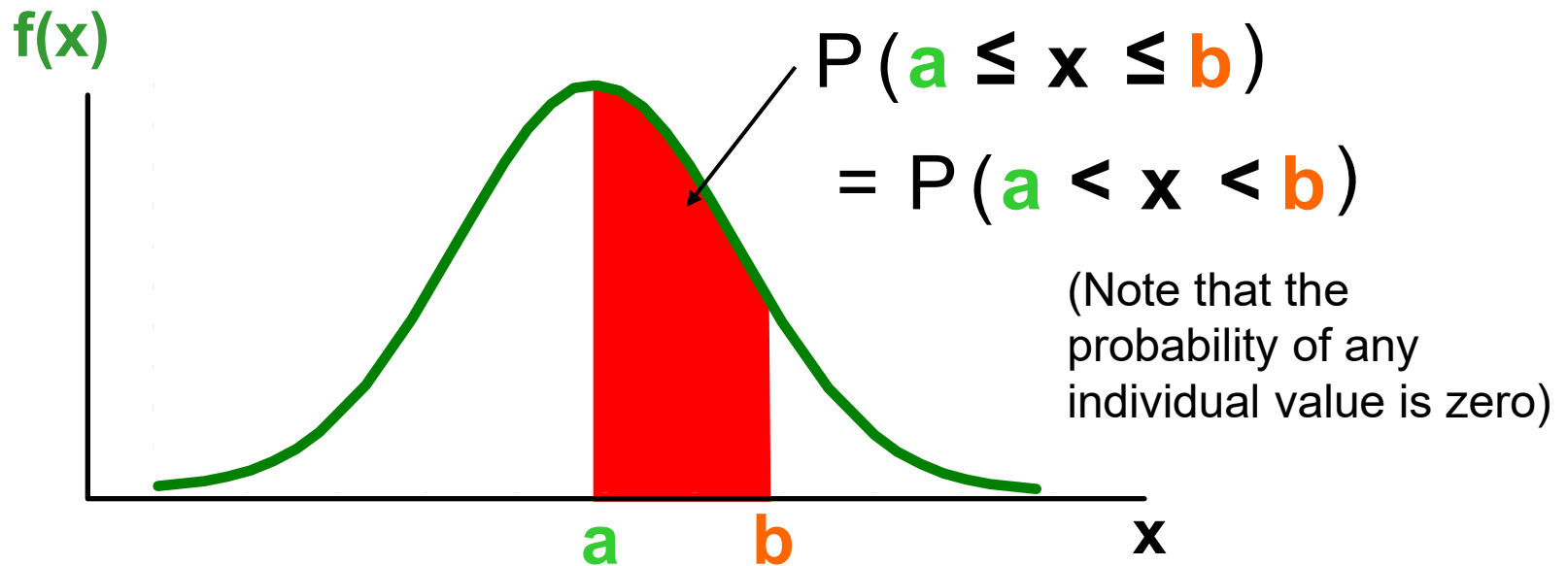
$$F(x_0) = \int_{x_m}^{x_0} f(x) dx$$

where  $x_m$  is the minimum value of the random variable  $X$



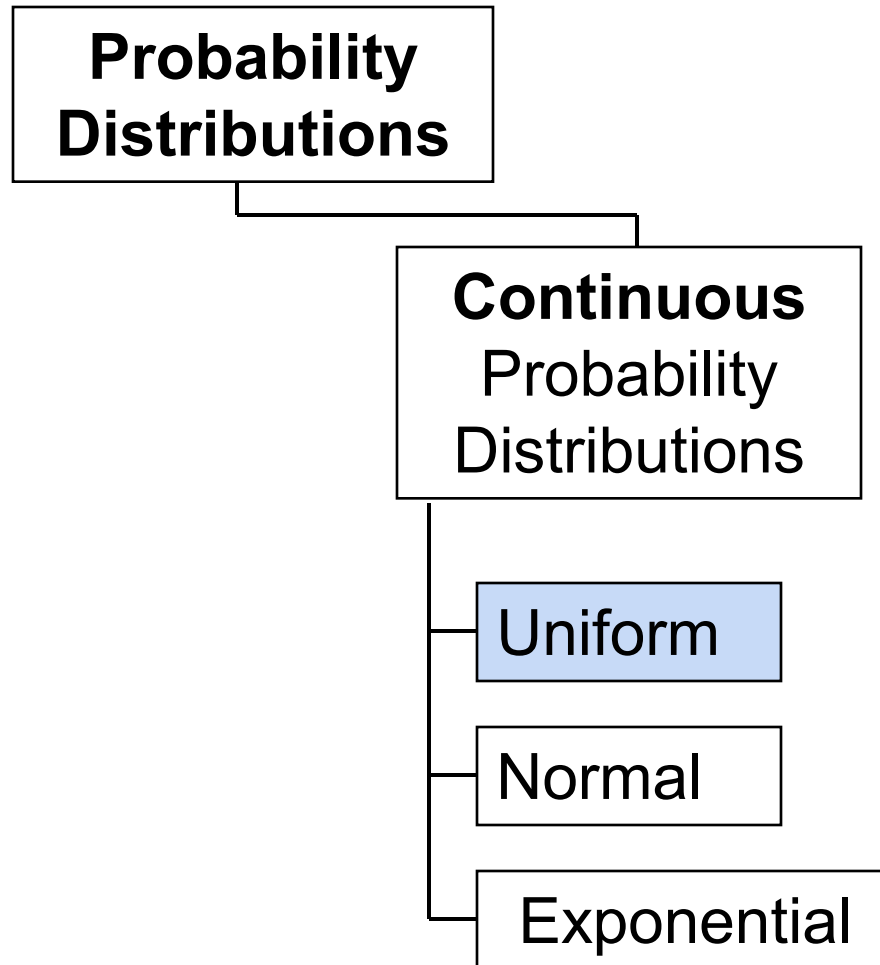
# Probability as an Area

Shaded **area under the curve** is the probability that  $X$  is between  $a$  and  $b$





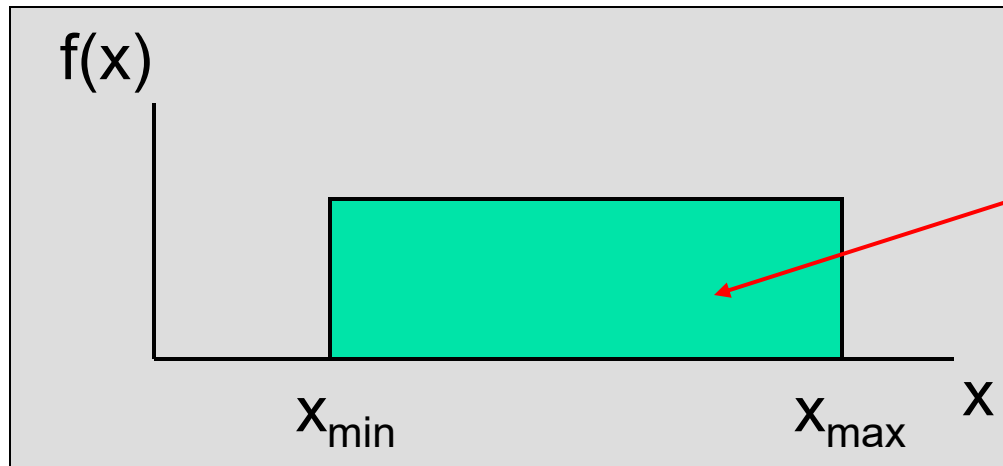
# The Uniform Distribution





# The Uniform Distribution

- The **uniform distribution** is a probability distribution that has **equal probabilities** for all possible outcomes of the random variable



Total area under the uniform probability density function is 1.0



# The Uniform Distribution

*(continued)*

The Continuous Uniform Distribution:

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

where

$f(x)$  = value of the density function at any  $x$  value

$a$  = minimum value of  $x$

$b$  = maximum value of  $x$



# Properties of the Uniform Distribution

- The **mean** of a uniform distribution is

$$\mu = \frac{a + b}{2}$$

- The **variance** is

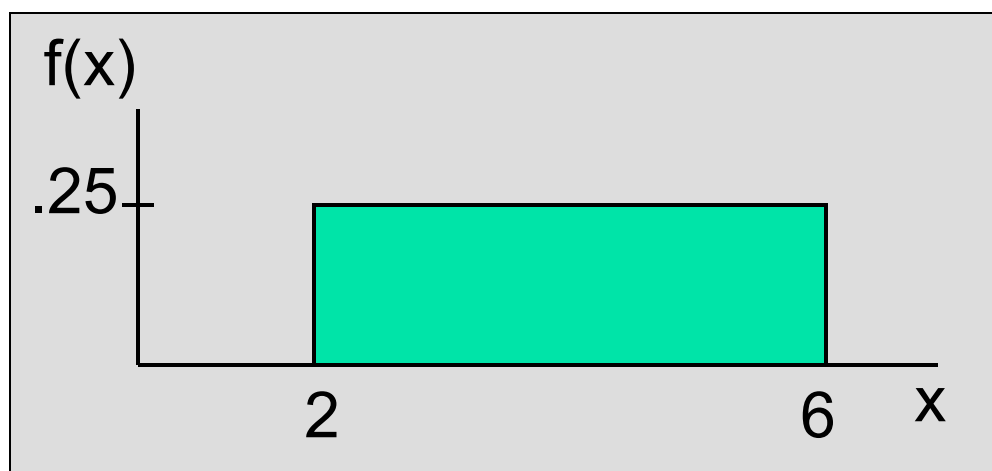
$$\sigma^2 = \frac{(b - a)^2}{12}$$



# Uniform Distribution Example

**Example:** Uniform probability distribution over the range  $2 \leq x \leq 6$ :

$$f(x) = \frac{1}{6 - 2} = .25 \quad \text{for } 2 \leq x \leq 6$$



$$\mu = \frac{a + b}{2} = \frac{2 + 6}{2} = 4$$

$$\sigma^2 = \frac{(b - a)^2}{12} = \frac{(6 - 2)^2}{12} = 1.333$$



# Expectations for Continuous Random Variables

- The mean of  $X$ , denoted  $\mu_X$ , is defined as the expected value of  $X$

$$\mu_X = E(X)$$

- The variance of  $X$ , denoted  $\sigma_X^2$ , is defined as the expectation of the squared deviation,  $(X - \mu_X)^2$ , of a random variable from its mean

$$\sigma_X^2 = E[(X - \mu_X)^2]$$



# Linear Functions of Variables

- Let  $W = a + bX$ , where  $X$  has mean  $\mu_X$  and variance  $\sigma_X^2$ , and  $a$  and  $b$  are constants

- Then the mean of  $W$  is

$$\mu_W = E(a + bX) = a + b\mu_X$$

- the variance is

$$\sigma_W^2 = \text{Var}(a + bX) = b^2\sigma_X^2$$

- the standard deviation of  $W$  is

$$\sigma_W = |b|\sigma_X$$



# Linear Functions of Variables

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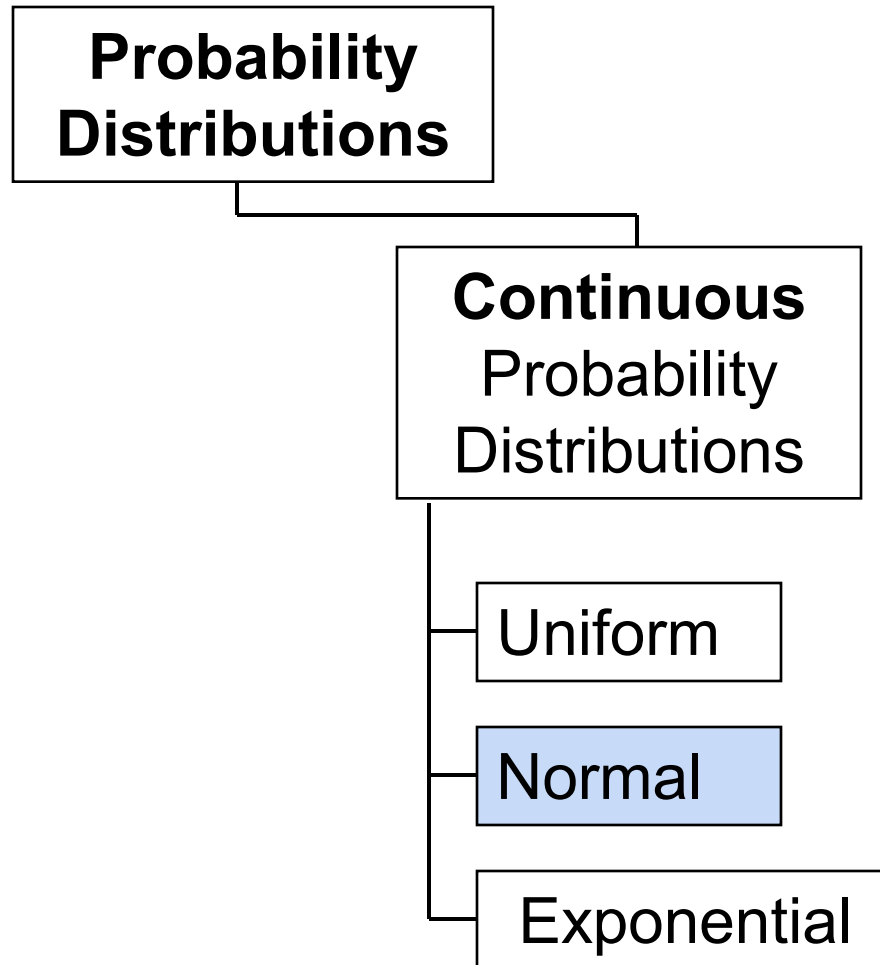
- An important special case of the previous results is the **standardized random variable**

$$Z = \frac{X - \mu_X}{\sigma_X}$$

- which has a mean 0 and variance 1



# The Normal Distribution





# The Normal Distribution

(continued)

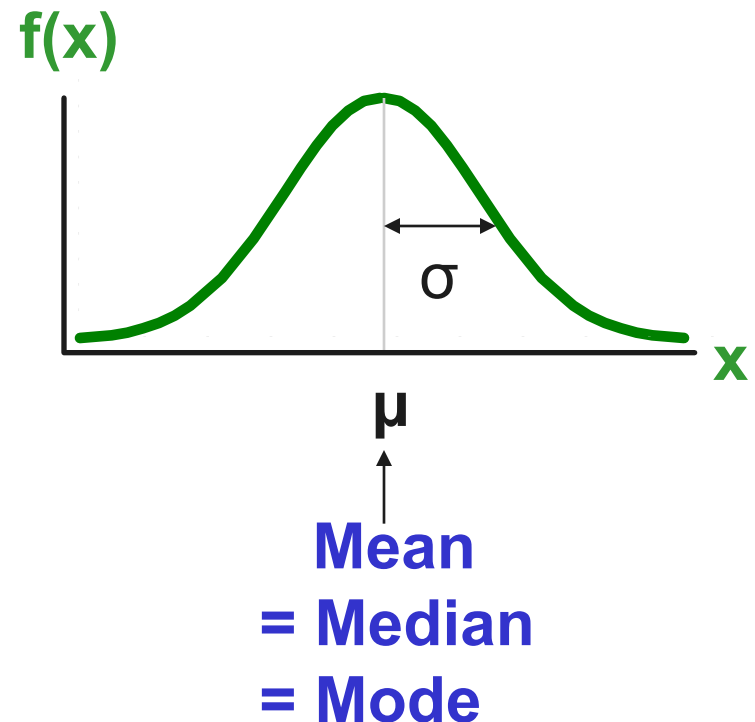
- Bell Shaped
- Symmetrical
- Mean, Median and Mode are Equal

Location is determined by the mean,  $\mu$

Spread is determined by the standard deviation,  $\sigma$

The random variable has an infinite theoretical range:

$+\infty$  to  $-\infty$





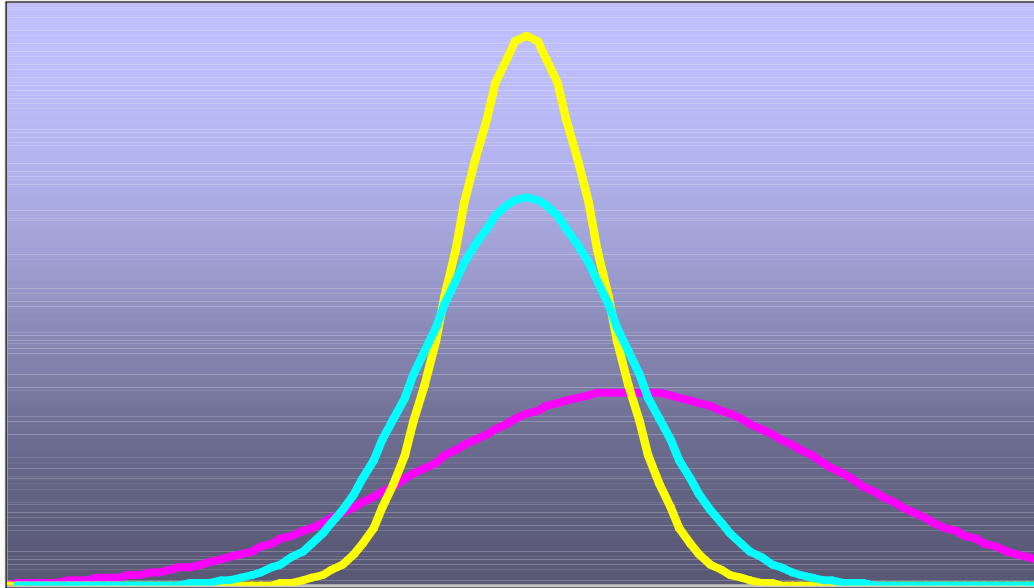
# The Normal Distribution

*(continued)*

- The normal distribution closely approximates the probability distributions of a wide range of random variables
- Distributions of sample means approach a normal distribution given a “large” sample size
- Computations of probabilities are direct and elegant
- The normal probability distribution has led to good business decisions for a number of applications



# Many Normal Distributions

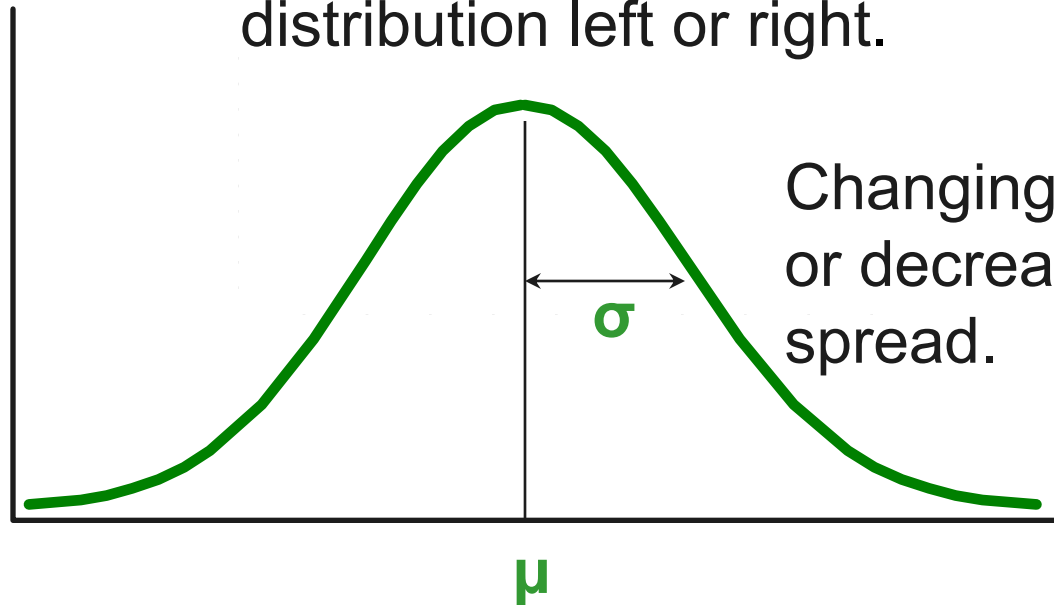


**By varying the parameters  $\mu$  and  $\sigma$ , we obtain different normal distributions**

# The Normal Distribution Shape

$f(x)$

Changing  $\mu$  shifts the distribution left or right.



Changing  $\sigma$  increases or decreases the spread.

Given the mean  $\mu$  and variance  $\sigma$  we define the normal distribution using the notation

$$X \sim N(\mu, \sigma^2)$$



# The Normal Probability Density Function

- The formula for the normal probability density function is

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2 / 2\sigma^2}$$

Where  $e$  = the mathematical constant approximated by 2.71828

$\pi$  = the mathematical constant approximated by 3.14159

$\mu$  = the population mean

$\sigma$  = the population standard deviation

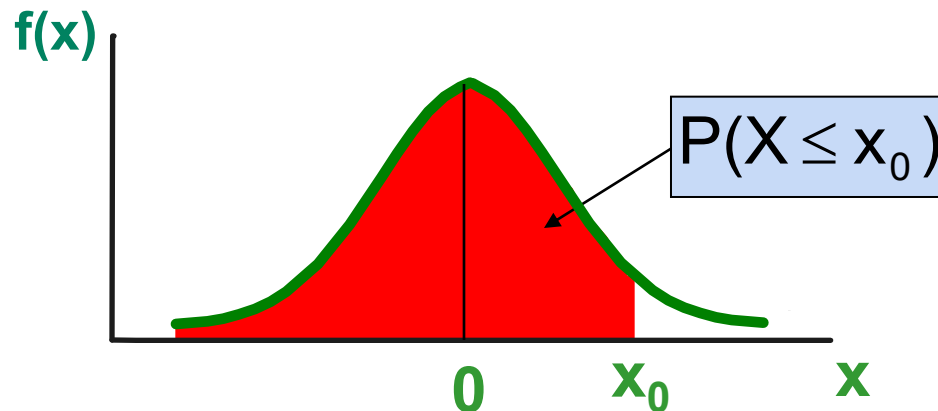
$x$  = any value of the continuous variable,  $-\infty < x < \infty$



# Cumulative Normal Distribution

- For a normal random variable  $X$  with mean  $\mu$  and variance  $\sigma^2$ , i.e.,  $X \sim N(\mu, \sigma^2)$ , the **cumulative distribution function** is

$$F(x_0) = P(X \leq x_0)$$

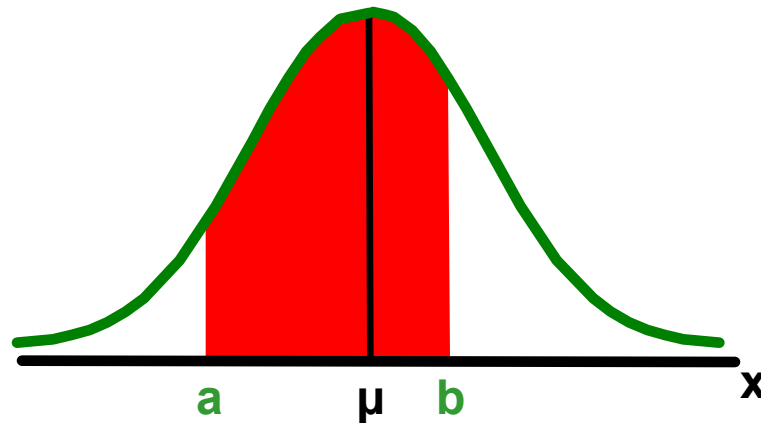




# Finding Normal Probabilities

The probability for a range of values is measured by the area under the curve

$$P(a < X < b) = F(b) - F(a)$$

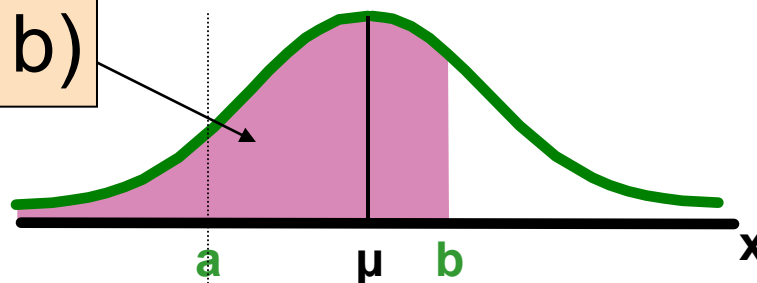




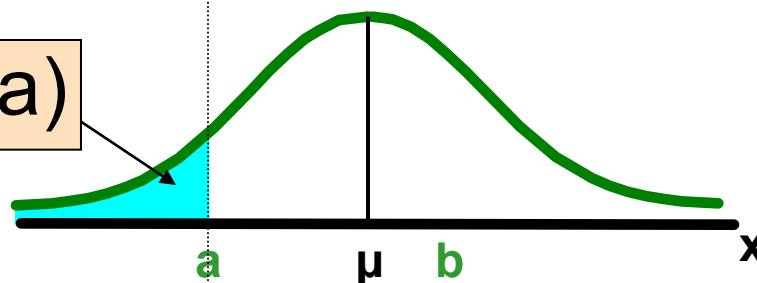
# Finding Normal Probabilities

(continued)

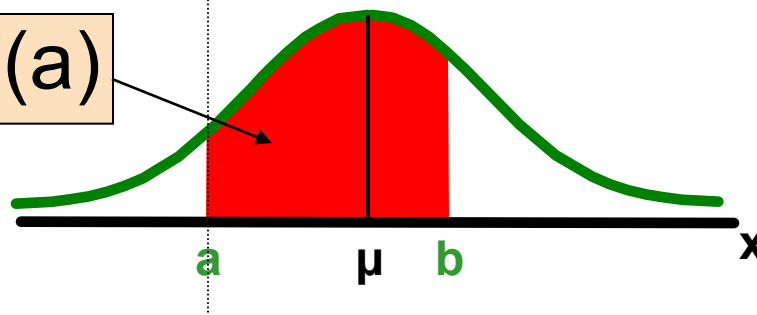
$$F(b) = P(X < b)$$



$$F(a) = P(X < a)$$



$$P(a < X < b) = F(b) - F(a)$$

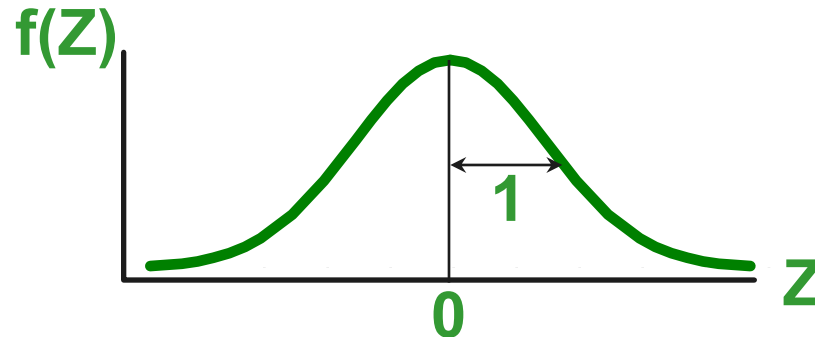




# The Standardized Normal

- Any normal distribution (with any mean and variance combination) can be transformed into the standardized normal distribution ( $Z$ ), with mean 0 and variance 1

$$Z \sim N(0,1)$$



- Need to transform  $X$  units into  $Z$  units by subtracting the mean of  $X$  and dividing by its standard deviation

$$Z = \frac{X - \mu}{\sigma}$$



# Example

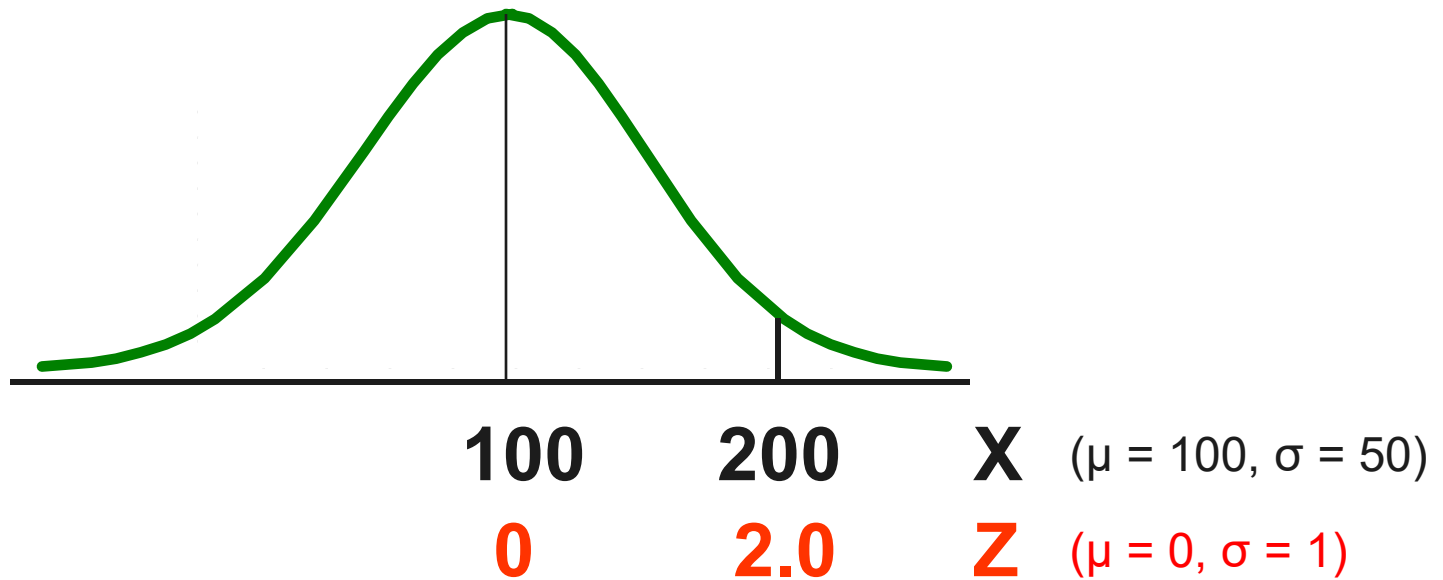
- If  $X$  is distributed normally with mean of 100 and standard deviation of 50, the  $Z$  value for  $X = 200$  is

$$Z = \frac{X - \mu}{\sigma} = \frac{200 - 100}{50} = 2.0$$

- This says that  $X = 200$  is two standard deviations (2 increments of 50 units) above the mean of 100.



# Comparing X and Z units

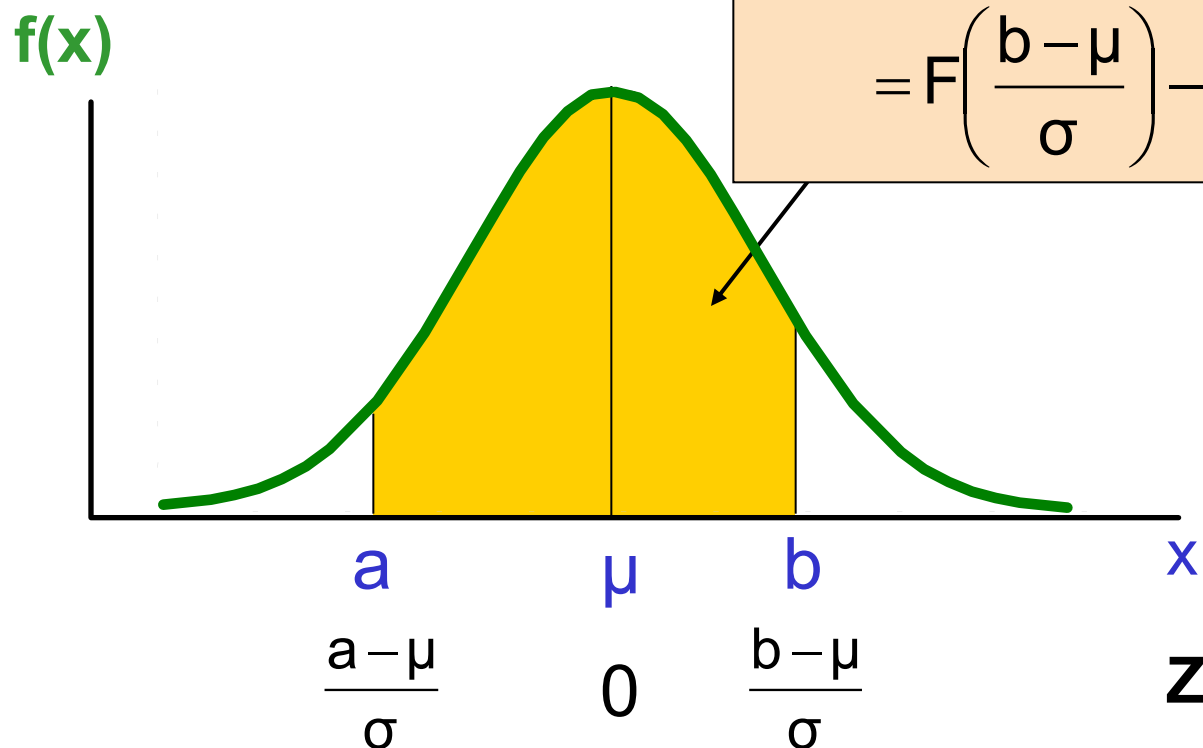


**Note that the distribution is the same, only the scale has changed. We can express the problem in original units (X) or in standardized units (Z)**



# Finding Normal Probabilities

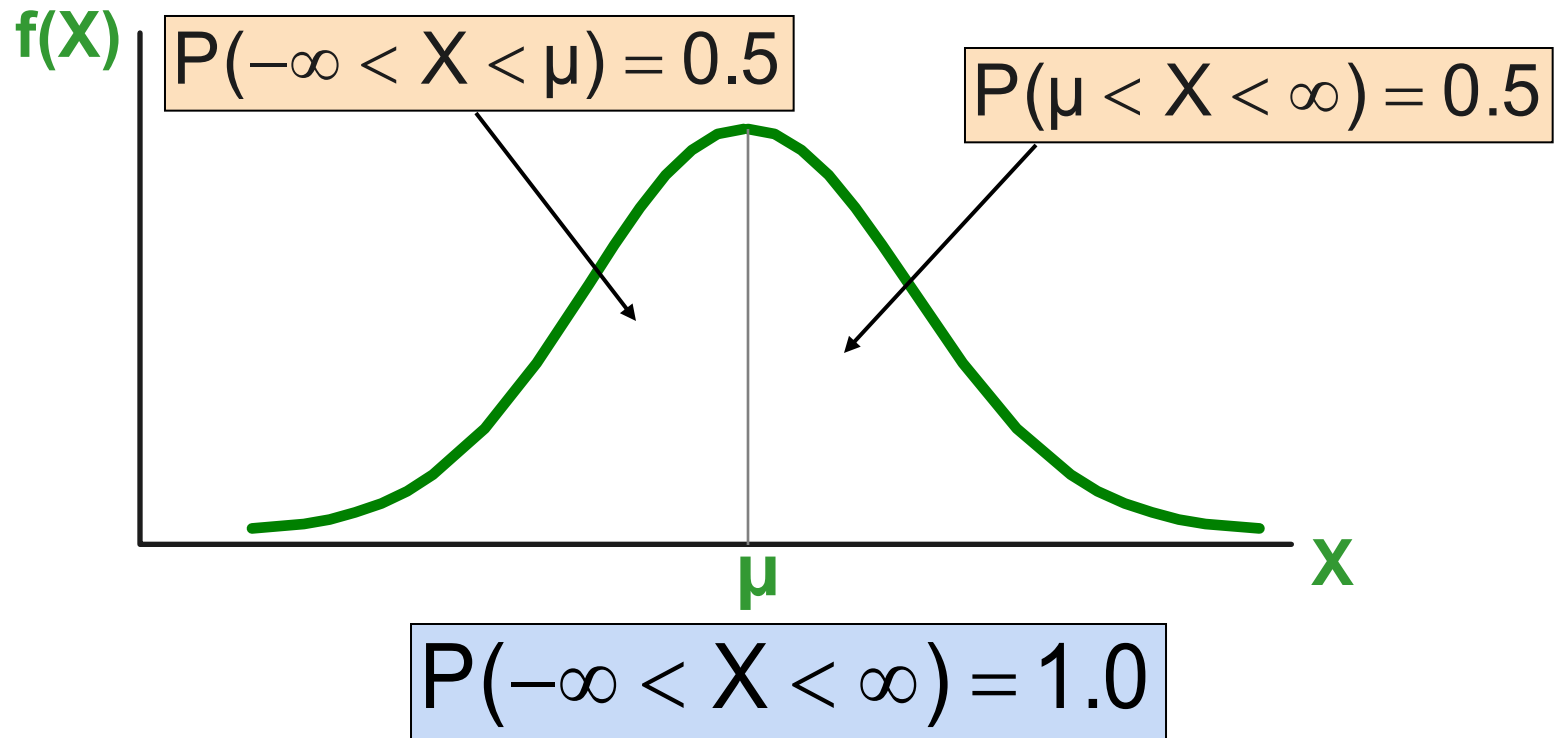
$$\begin{aligned} P(a < X < b) &= P\left(\frac{a - \mu}{\sigma} < Z < \frac{b - \mu}{\sigma}\right) \\ &= F\left(\frac{b - \mu}{\sigma}\right) - F\left(\frac{a - \mu}{\sigma}\right) \end{aligned}$$





# Probability as Area Under the Curve

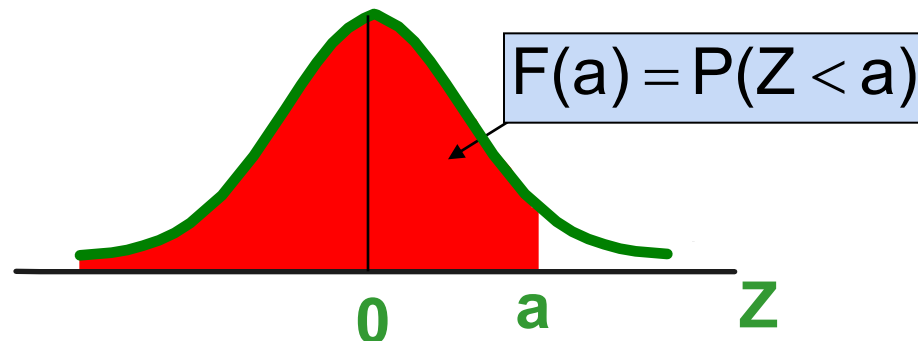
The total area under the curve is 1.0, and the curve is symmetric, so half is above the mean, half is below





# Appendix Table 1

- The Standardized Normal table in the textbook (Appendix Table 1) shows values of the cumulative normal distribution function
- For a given Z-value  $a$ , the table shows  $F(a)$  (the area under the curve from negative infinity to  $a$ )



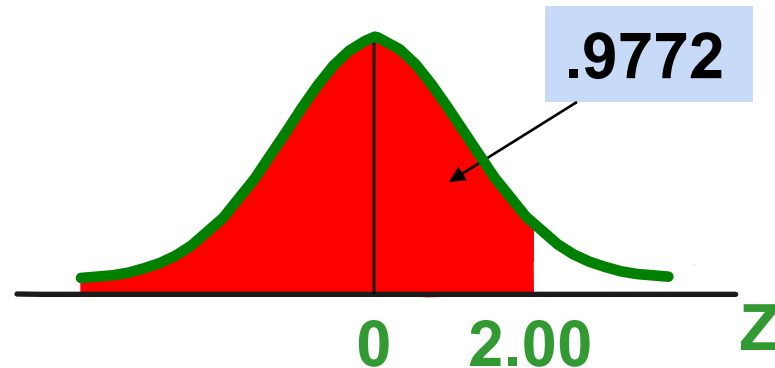


# The Standardized Normal Table

- Appendix Table 1 gives the probability  $F(a)$  for any value  $a$

Example:

$$P(Z < 2.00) = .9772$$





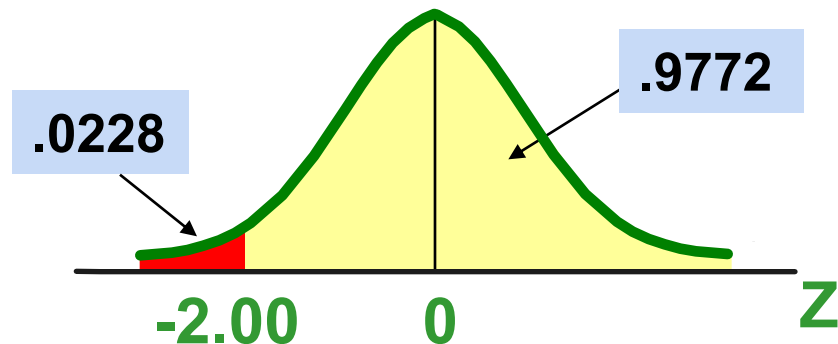
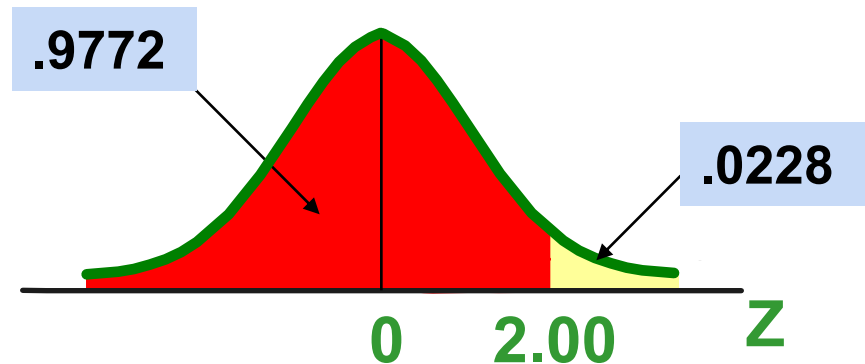
# The Standardized Normal Table

*(continued)*

- For **negative Z-values**, use the fact that the distribution is symmetric to find the needed probability:

**Example:**

$$\begin{aligned} P(Z < -2.00) &= 1 - 0.9772 \\ &= 0.0228 \end{aligned}$$





# General Procedure for Finding Probabilities

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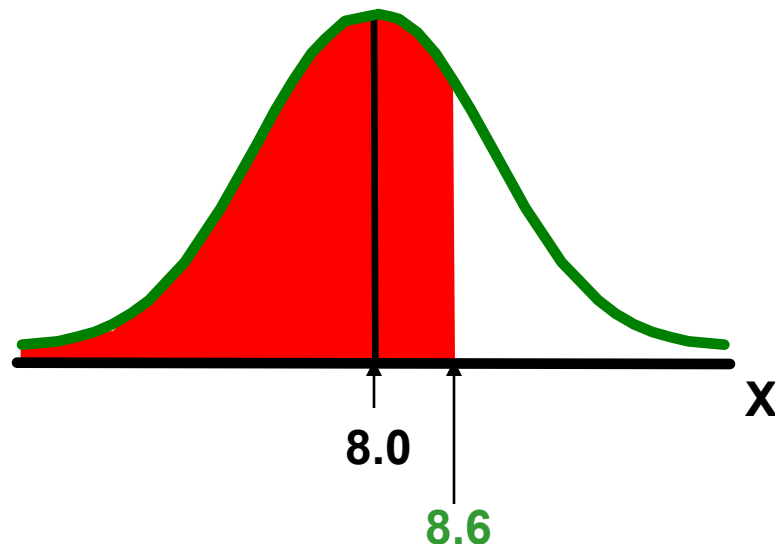
To find  $P(a < X < b)$  when  $X$  is distributed normally:

- Draw the normal curve for the problem in terms of  $X$
- Translate  $X$ -values to  $Z$ -values
- Use the Cumulative Normal Table



# Finding Normal Probabilities

- Suppose  $X$  is normal with mean 8.0 and standard deviation 5.0
- Find  $P(X < 8.6)$



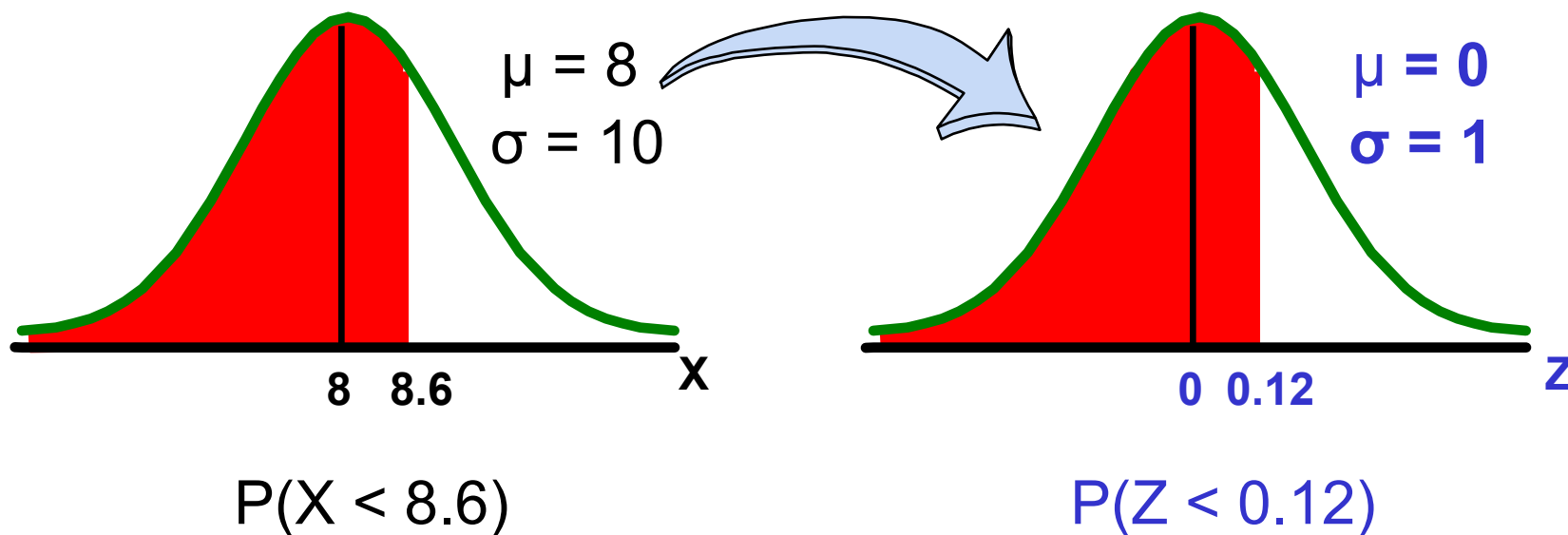


# Finding Normal Probabilities

(continued)

- Suppose  $X$  is normal with mean 8.0 and standard deviation 5.0. Find  $P(X < 8.6)$

$$Z = \frac{X - \mu}{\sigma} = \frac{8.6 - 8.0}{5.0} = 0.12$$

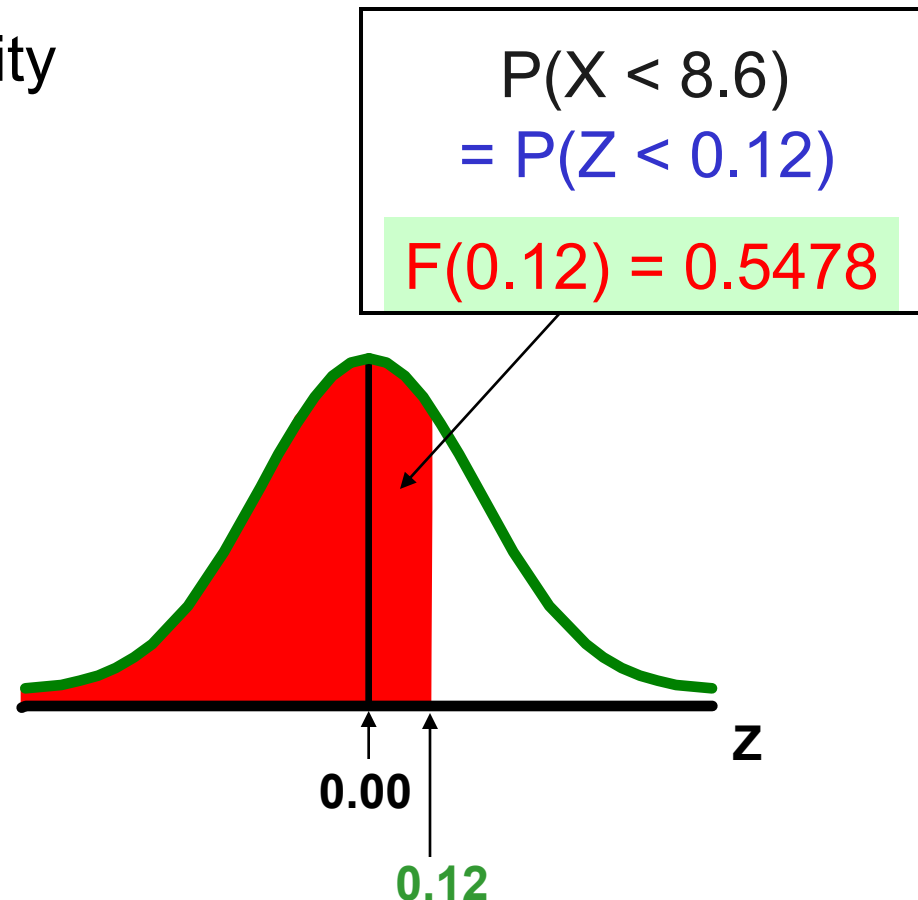




# Solution: Finding $P(Z < 0.12)$

Standardized Normal Probability  
Table (Portion)

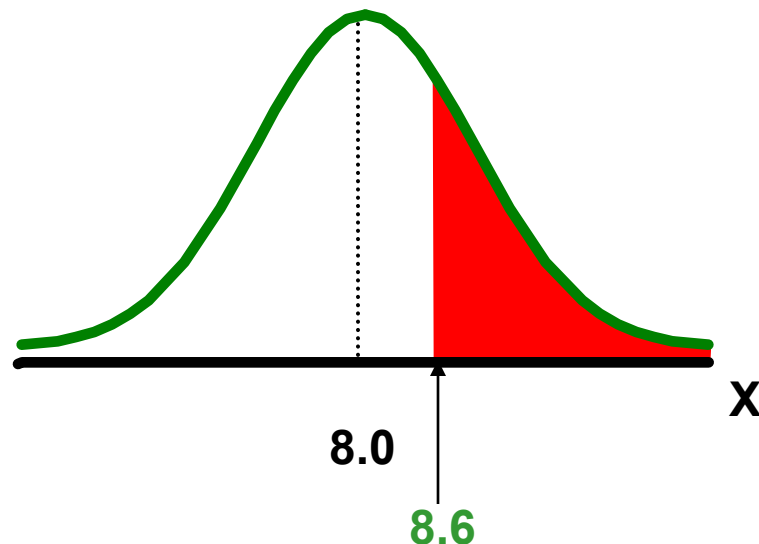
| z   | F(z)  |
|-----|-------|
| .10 | .5398 |
| .11 | .5438 |
| .12 | .5478 |
| .13 | .5517 |





# Upper Tail Probabilities

- Suppose  $X$  is normal with mean 8.0 and standard deviation 5.0.
- Now Find  $P(X > 8.6)$

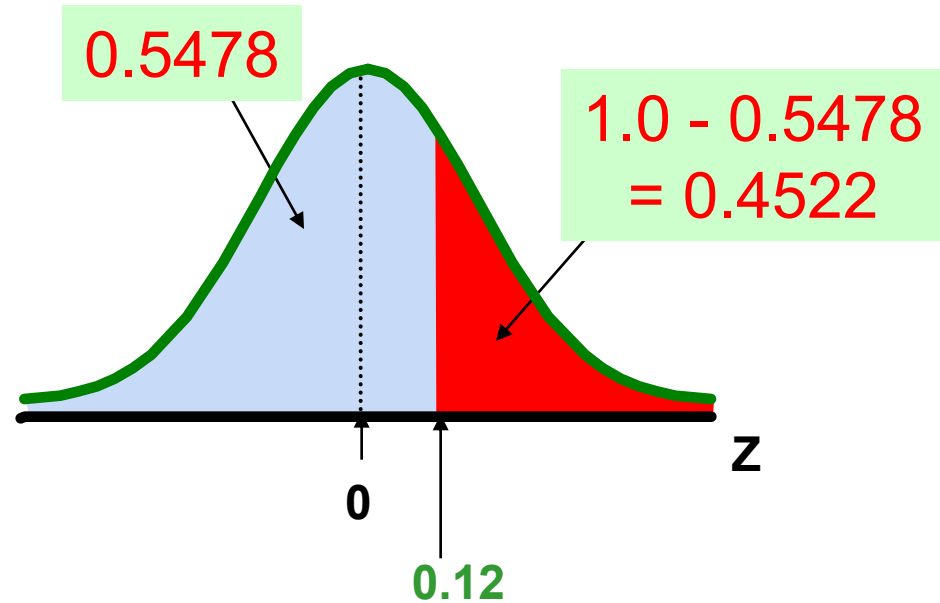
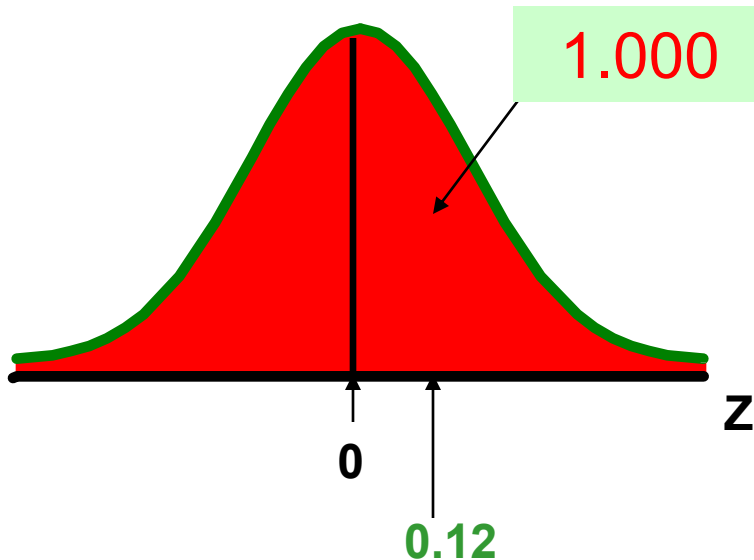


# Upper Tail Probabilities

(continued)

- Now Find  $P(X > 8.6)$ ...

$$\begin{aligned} P(X > 8.6) &= P(Z > 0.12) = 1.0 - P(Z \leq 0.12) \\ &= 1.0 - 0.5478 = 0.4522 \end{aligned}$$





# Finding the X value for a Known Probability

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- Steps to find the X value for a known probability:
  1. Find the Z value for the known probability
  2. Convert to X units using the formula:

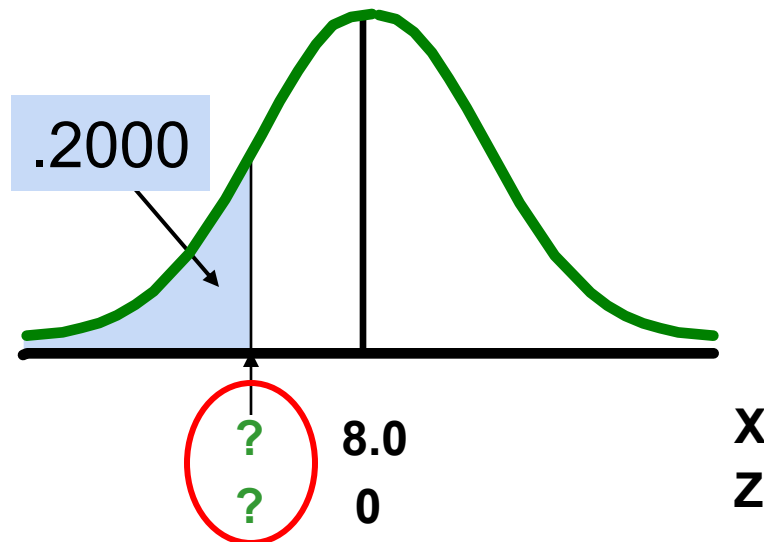
$$X = \mu + Z\sigma$$

# Finding the $X$ value for a Known Probability

(continued)

Example:

- Suppose  $X$  is normal with mean 8.0 and standard deviation 5.0.
- Now find the  $X$  value so that only 20% of all values are below this  $X$



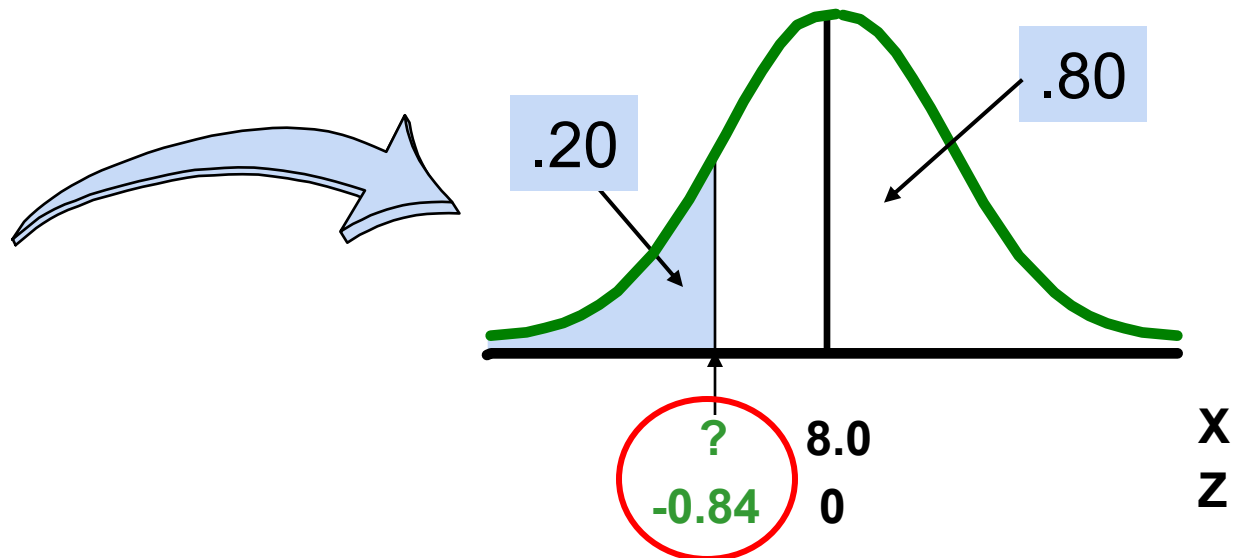
# Find the Z value for 20% in the Lower Tail

1. Find the Z value for the known probability

Standardized Normal Probability Table (Portion)

| z   | F(z)  |
|-----|-------|
| .82 | .7939 |
| .83 | .7967 |
| .84 | .7995 |
| .85 | .8023 |

- 20% area in the lower tail is consistent with a Z value of **-0.84**





# Finding the X value

2. Convert to X units using the formula:

$$\begin{aligned} X &= \mu + Z\sigma \\ &= 8.0 + (-0.84)5.0 \\ &= 3.80 \end{aligned}$$

So 20% of the values from a distribution with mean 8.0 and standard deviation 5.0 are less than 3.80



# Assessing Normality

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- Not all continuous random variables are normally distributed
- It is important to evaluate how well the data is approximated by a normal distribution



# The Normal Probability Plot

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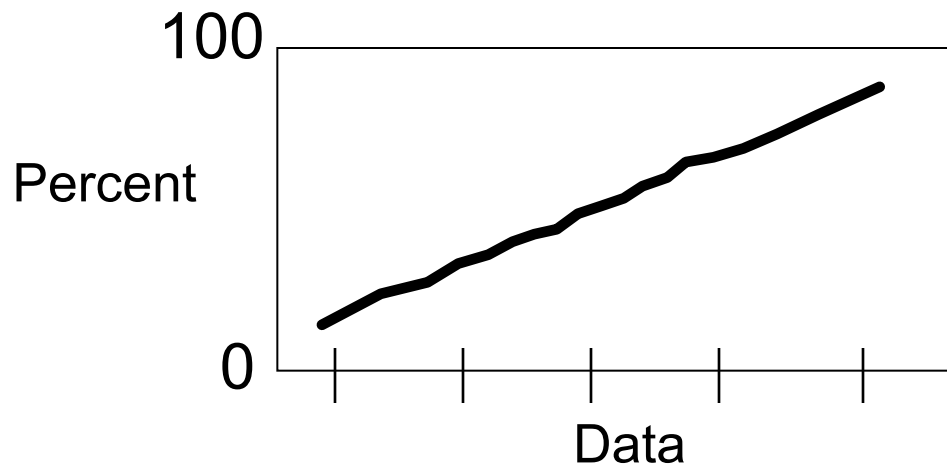
- Normal probability plot
  - Arrange data from low to high values
  - Find cumulative normal probabilities for all values
  - Examine a plot of the observed values vs. cumulative probabilities (with the cumulative normal probability on the vertical axis and the observed data values on the horizontal axis)
  - Evaluate the plot for evidence of linearity



# The Normal Probability Plot

*(continued)*

A normal probability plot for data from a normal distribution will be approximately linear:

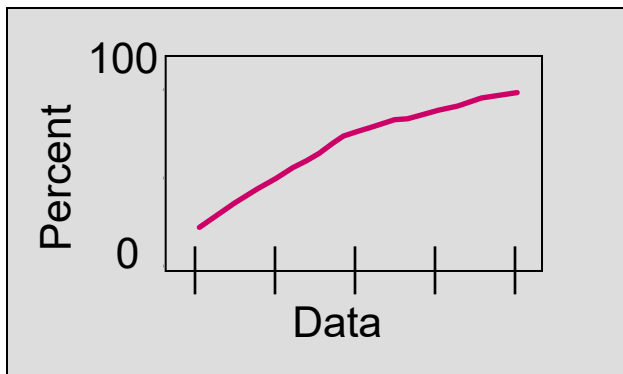




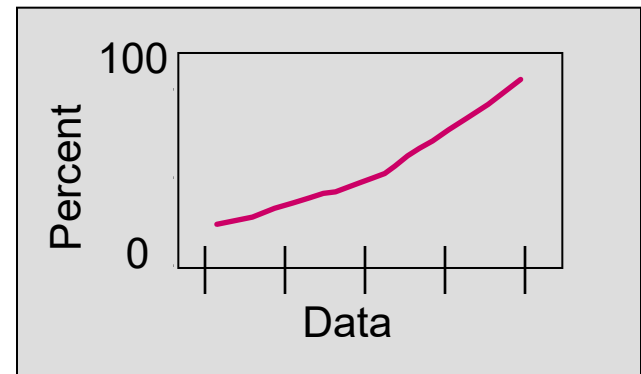
# The Normal Probability Plot

*(continued)*

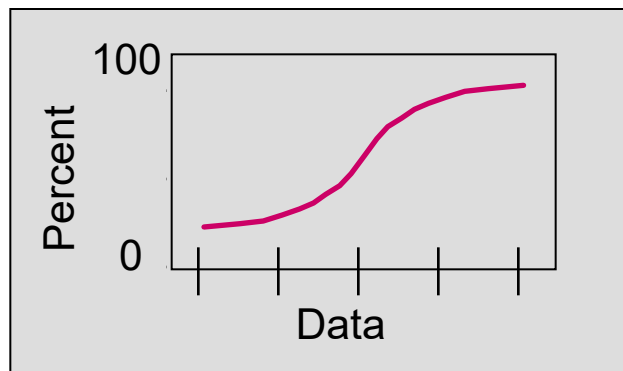
## Left-Skewed



## Right-Skewed



## Uniform



Nonlinear plots  
indicate a deviation  
from normality



# Normal Distribution Approximation for Binomial Distribution

- Recall the binomial distribution:
  - n independent trials
  - probability of success on any given trial = P
- Random variable X:
  - $X_i = 1$  if the  $i^{\text{th}}$  trial is “success”
  - $X_i = 0$  if the  $i^{\text{th}}$  trial is “failure”

$$E(X) = \mu = nP$$

$$\text{Var}(X) = \sigma^2 = nP(1-P)$$



# Normal Distribution Approximation for Binomial Distribution

*(continued)*

- The shape of the binomial distribution is **approximately normal** if  $n$  is large
- The normal is a good approximation to the binomial when  **$nP(1 - P) > 9$**
- Standardize to  $Z$  from a binomial distribution:

$$Z = \frac{X - E(X)}{\sqrt{\text{Var}(X)}} = \frac{X - np}{\sqrt{nP(1-P)}}$$



# Normal Distribution Approximation for Binomial Distribution

*(continued)*

- Let  $X$  be the number of successes from  $n$  independent trials, each with probability of success  $P$ .
- If  $nP(1 - P) > 9$ ,

$$P(a < X < b) = P\left(\frac{a - nP}{\sqrt{nP(1-P)}} \leq Z \leq \frac{b - nP}{\sqrt{nP(1-P)}}\right)$$



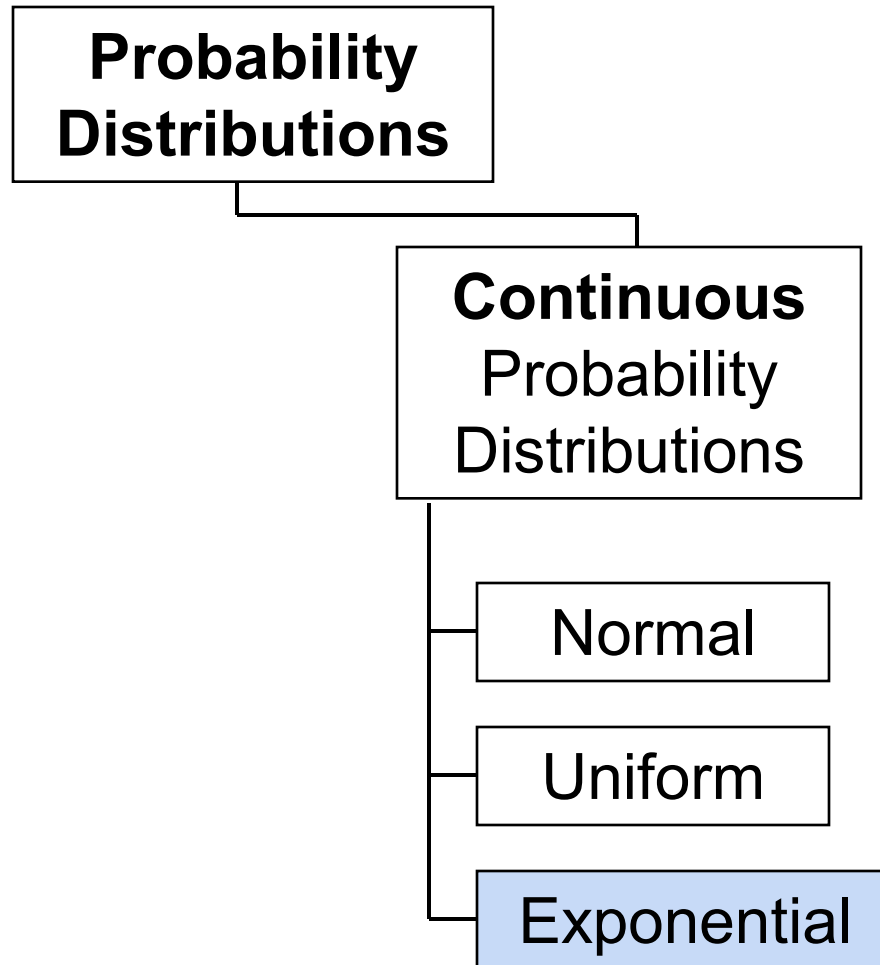
# Binomial Approximation Example

- 40% of all voters support ballot proposition A. What is the probability that between 76 and 80 voters indicate support in a sample of  $n = 200$  ?
  - $E(X) = \mu = nP = 200(0.40) = 80$
  - $\text{Var}(X) = \sigma^2 = nP(1 - P) = 200(0.40)(1 - 0.40) = 48$   
( note:  $nP(1 - P) = 48 > 9$  )

$$\begin{aligned} P(76 < X < 80) &= P\left( \frac{76 - 80}{\sqrt{200(0.4)(1-0.4)}} \leq Z \leq \frac{80 - 80}{\sqrt{200(0.4)(1-0.4)}} \right) \\ &= P(-0.58 < Z < 0) \\ &= F(0) - F(-0.58) \\ &= 0.5000 - 0.2810 = 0.2190 \end{aligned}$$



# The Exponential Distribution





# The Exponential Distribution

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- Used to model the **length of time between two occurrences** of an event (the time between arrivals)
  
- Examples:
  - Time between trucks arriving at an unloading dock
  - Time between transactions at an ATM Machine
  - Time between phone calls to the main operator



# The Exponential Distribution

*(continued)*

- The **exponential random variable**  $T$  ( $t > 0$ ) has a probability density function

$$f(t) = \lambda e^{-\lambda t} \quad \text{for } t > 0$$

- Where
  - $\lambda$  is the mean number of occurrences per unit time
  - $t$  is the number of time units until the next occurrence
  - $e = 2.71828$
- $T$  is said to follow an exponential probability distribution



# The Exponential Distribution

- Defined by a single parameter, its mean  $\lambda$  (lambda)
- The **cumulative distribution function** (the probability that an arrival time is less than some specified time  $t$ ) is

$$F(t) = 1 - e^{-\lambda t}$$

where  $e$  = mathematical constant approximated by 2.71828  
 $\lambda$  = the population mean number of arrivals per unit  
 $t$  = any value of the continuous variable where  $t > 0$



# Exponential Distribution Example

**Example:** Customers arrive at the service counter at the rate of 15 per hour. What is the probability that the arrival time between consecutive customers is less than three minutes?

- The mean number of arrivals per hour is 15, so  $\lambda = 15$
- Three minutes is .05 hours
- $P(\text{arrival time} < .05) = 1 - e^{-\lambda X} = 1 - e^{-(15)(.05)} = 0.5276$
- So there is a 52.76% probability that the arrival time between successive customers is less than three minutes



# Summary

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- Defined continuous random variables
- Presented key continuous probability distributions and their properties
  - uniform, normal, exponential
- Found probabilities using formulas and tables
- Interpreted normal probability plots
- Examined when to apply different distributions
- Applied the normal approximation to the binomial distribution